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Two Relaxed CQ Methods for the Split Feasibility Problem with Multiple Output Sets

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Abstract
In this paper, two relaxed CQ algorithms with non-inertial and inertial steps are proposed for solving the split feasibility problems with multiple output sets (SFPMOS) in infinite-dimensional real Hilbert spaces. The step size is determined dynamically without requiring prior information about the operator norm. Furthermore, the proposed algorithms are proven to converge strongly to the minimum-norm solution of the SFPMOS. Some applications of our main results regarding the solution of the split feasibility problem are presented. Finally, we give two numerical examples to illustrate the efficiency and implementation of our algorithms in comparison with existing algorithms in the literature.

Keywords Split feasibility problems · Multiple output sets · Relaxed CQ algorithm · Self-adaptive · Strong convergence

Mathematics Subject Classification 47J20 · 47J25 · 49N45 · 65J15

1 Introduction

The split feasibility problem (SFP) was initially introduced by Censor and Elfving [5] in 1994 to model inverse problems in finite-dimensional Hilbert spaces and afterward extended to infinite dimensional spaces as well as applied successfully in the field

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of intensity-modulated radiation therapy treatment planning, see [5–8]. The SFP is formulated as follows.

$$\text{Find } u^* \in \mathcal{C} \text{ such that } \mathcal{F}u^* \in \mathcal{Q}, \tag{1}$$

where $\mathcal{C}$ and $\mathcal{Q}$ are nonempty closed convex subsets of real Hilbert spaces $\mathcal{H}$ and $\mathcal{H}_1$, respectively, and $\mathcal{F} : \mathcal{H} \to \mathcal{H}_1$ is a bounded linear operator. The most popular and practical algorithm for solving the SFP is Byrne's CQ algorithm [3, 4], which is defined as follows:

$$x^{k+1} = P_{\mathcal{C}}\big(x^k - \gamma \mathcal{F}^*(I - P_{\mathcal{Q}})\mathcal{F}x^k\big), \tag{2}$$

where I is the identity mapping in $\mathcal{H}_1$, $P_{\mathcal{C}}$ and $P_{\mathcal{Q}}$ are the metric projections onto $\mathcal{C}$ and $\mathcal{Q}$, respectively, $\mathcal{F}^*$ is the adjoint of $\mathcal{F}$, and $\gamma \in \big(0, 2/\|\mathcal{F}\|^2\big)$. It is worth noting that Byrne's CQ algorithm is a special case of the gradient-projection method in constrained convex minimization problems. Xu [17] proved the weak convergence of (2) in the frameworks of infinite-dimensional Hilbert space. In order to obtain strong convergence, in the same year, Wang and Xu [15] introduced an iterative algorithm as follows:

$$x^{k+1} = P_{\mathcal{C}}\big[(1 - \alpha_k)\big(x^k - \gamma \mathcal{F}^*(I - P_{\mathcal{Q}})\mathcal{F}x^k\big)\big], \tag{3}$$

where $\gamma \in \big(0, 2/\|\mathcal{F}\|^2\big)$ and the sequence $\{\alpha_k\}$ satisfies the following conditions

$$\alpha_k \in (0, 1) \text{ for all } k, \ \lim_{k\to\infty} \alpha_k = 0, \ \sum_{k=0}^{\infty} \alpha_k = \infty, \tag{4}$$

$$\text{either } \sum_{k=1}^{\infty} |\alpha_k - \alpha_{k-1}| < \infty \text{ or } \lim_{k\to\infty} \frac{|\alpha_k - \alpha_{k-1}|}{\alpha_k} = 0. \tag{5}$$

They proved that the sequence $\{x^k\}$ generated by (3) converges strongly to the minimum-norm solution of the SFP. Yu et al. [20] proved that the sequence $\{x^k\}$ generated by (3) is strongly convergent without the condition (5).

The computation of a projection onto a closed convex subset is generally difficult. To overcome this difficulty, Fukushima [11] suggested a way to calculate the projection onto a level set of a convex function by computing a sequence of projections onto half-spaces containing the original level set. This idea is followed by Yang [18], who introduced a relaxed CQ algorithm in a finite-dimensional Hilbert space as follows:

$$x^{k+1} = P_{\mathcal{C}_k}\big(x^k - \gamma \mathcal{F}^*(I - P_{\mathcal{Q}_k})\mathcal{F}x^k\big), \tag{6}$$

where $\gamma \in \big(0, 2/\|\mathcal{F}\|^2\big)$, $\mathcal{C}_k$ and $\mathcal{Q}_k$ are half-spaces. More importantly, since the projections onto half-spaces $\mathcal{C}_k$ and $\mathcal{Q}_k$ have closed forms, the relaxed CQ algorithm (6) is implementable. However, to implement the relaxed CQ algorithm, one has to calculate or estimate the operator norm $\|\mathcal{F}\|$, which is in general not an easy task in practice. To overcome this difficulty, López et al. [13] improved Yang's relaxed CQ algorithm as follows:

$$x^{k+1} = P_{\mathcal{C}_k}\big(x^k - \gamma_k \mathcal{F}^*(I - P_{\mathcal{Q}_k})\mathcal{F}x^k\big), \tag{7}$$

where

$$\gamma_k = \rho_k \frac{\|(I - P_{\mathcal{Q}_k})\mathcal{F}x^k\|^2}{\|\mathcal{F}^*(I - P_{\mathcal{Q}_k})\mathcal{F}x^k\|^2}, \tag{8}$$

with ρ_k satisfies the following condition

$$\rho_k \in (0, 2) \text{ for all } k, \ \inf_{k \geq 1} \rho_k(2 - \rho_k) > 0. \tag{9}$$

Very recently, Yu and Wang [19] proposed several new relaxed CQ algorithms. The main idea of the algorithms is to use the current half-spaces $\mathcal{C}_k$ and $\mathcal{Q}_k$, and the half-spaces $\mathcal{C}_{k-1}$ and $\mathcal{Q}_{k-1}$ constructed in the previous steps. One of their algorithms is as follows:

$$x^{k+1} = P_{\mathcal{C}_k} P_{\mathcal{C}_{k-1}}\big[x^k - \gamma_k\big(\mu \mathcal{F}^*(I - P_{\mathcal{Q}_k})\mathcal{F}x^k + (1-\mu)\mathcal{F}^*(I - P_{\mathcal{Q}_{k-1}})\mathcal{F}x^k\big)\big], \tag{10}$$

where $x^0, x^1 \in \mathcal{H}$, $\mu \in (0, 1)$, and

$$\gamma_k = \rho_k \frac{\mu\|(I - P_{\mathcal{Q}_k})\mathcal{F}x^k\|^2 + (1-\mu)\|(I - P_{\mathcal{Q}_{k-1}})\mathcal{F}x^k\|^2}{\mu\|\mathcal{F}^*(I - P_{\mathcal{Q}_k})\mathcal{F}x^k\|^2 + (1-\mu)\|\mathcal{F}^*(I - P_{\mathcal{Q}_{k-1}})\mathcal{F}x^k\|^2}, \tag{11}$$

with ρ_k satisfies the condition (9). The weak convergence of the (10) is proved [19].

In the literature, there are various generalizations of SFP, one of which is the Split Feasibility Problem with Multiple Output Sets (SFPMOS) [9, 14, 16]. For each $j = 1, \ldots, N$, let $\mathcal{H}_j$ be a real Hilbert space, $\mathcal{Q}_j \subseteq \mathcal{H}_j$ a nonempty closed convex subset, and $\mathcal{F}_j : \mathcal{H} \to \mathcal{H}_j$ a bounded linear operator. The SFPMOS can be formulated as follows.

$$\text{Find } u^* \in \mathcal{C} \text{ such that } \mathcal{F}_j u^* \in \mathcal{Q}_j \quad \forall j = 1, \ldots, N. \tag{12}$$

In what follows we will denote by Ω the solution set of the problem SFPMOS (12), that is, $\Omega = \{u^* \in \mathcal{C} \mid \mathcal{F}_j u^* \in \mathcal{Q}_j \ \forall j = 1, \ldots, N\}$. In [14], Reich et al. proposed the following algorithm

$$x^{k+1} = \alpha_k u + (1-\alpha_k) P_{\mathcal{C}}\Big[x^k - \gamma_k \sum_{j=1}^{N} \mathcal{F}_j^*(I - P_{\mathcal{Q}_j})\mathcal{F}_j x^k\Big], \quad k \geq 0, \tag{13}$$

that can be used to find $u^* \in \Omega$ such that $u^* = P_\Omega u$, where α_k satisfies the condition (4) and

$$0 < \gamma_k < \frac{2}{N \max_{j=1,\dots,N}\{\|\mathcal{F}_j\|^2\}}. \tag{14}$$

Wang [16] introduced a step size sequence $\{\gamma_k\}$ for (13), when u replaced by $f(x^k)$, f is a contraction, as follows

$$\gamma_k = \begin{cases} \frac{\sum_{j=1}^N \|(I-P_{Q_j})\mathcal{F}_j x^k\|^2}{\|\sum_{j=1}^N \mathcal{F}_j^*(I-P_{Q_j})\mathcal{F}_j x^k\|^2}, & \text{if } \|\sum_{j=1}^N \mathcal{F}_j^*(I-P_{Q_j})\mathcal{F}_j x^k\| \neq 0, \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

Cuong et al. [9] introduced a self-adaptive step size algorithm for solving SFPMOS, where the step size γ_k is chosen in such a way that

$$\gamma_k = \begin{cases} \frac{\sum_{j=1}^N \|(I-P_{Q_j})\mathcal{F}_j x^k\|^2}{2\left(\sum_{j=1}^N \|\mathcal{F}_j^*(I-P_{Q_j})\mathcal{F}_j x^k\|\right)^2}, & \text{if } \sum_{j=1}^N \|\mathcal{F}_j^*(I-P_{Q_j})\mathcal{F}_j x^k\| \neq 0, \\ 0, & \text{otherwise.} \end{cases} \tag{16}$$

Motivated and inspired by the above-mentioned works, in the present manuscript, we consider two relaxed CQ algorithms with an inertial procedure and non-inertial for finding a minimum-norm solution to the SFPMOS (12) in real Hilbert spaces. We establish global convergence of our scheme under some easy-to-verify assumptions and without the prior knowledge of the norms of the operators $\mathcal{F}_j$ like the algorithm (13) of Reich et al [14]. Unlike the algorithms of Reich et al. [14], Wang [16], and Cuong et al. [9], our algorithms use the projection onto the half-spaces. Since the projection onto the half-space has closed form, the proposed algorithm is thus easy to be implemented. Furthermore, compared with the algorithms of Wang [16] and Cuong et al. [9], the step size we constructed is often larger, while a larger step size often means faster convergence speed. Some applications of our main results regarding the solution of SFP are presented.

The remaining part of this paper is organized as follows: The next section displays some lemmas that will be used for the validity and convergence of the algorithms. The third section is devoted to the description of our proposed algorithms and the strong convergence results. In Sect. 4, we consider some applications to the problem of finding the minimum-norm solution of the SFP. In Sect. 5, we illustrate the proposed method by considering two numerical experiments, one of which is numerical experiments in signal recovery problems.

2 Preliminaries

In this section, we introduce some mathematical symbols, definitions, and lemmas which can be used in the proof of our main result. Let $\mathcal{H}$ be a real Hilbert space with inner product $\langle ., . \rangle$ and norm $\|.\|$. In what follows, we write $x^k \rightharpoonup x$ to indicate that the sequence $\{x^k\}$ converges weakly to x while $x^k \to x$ indicate that the sequence $\{x^k\}$ converges strongly to x. Let $\omega_w(x^k)$ denotes the weak ω-limit set of $\{x^k\}$, that is, the set of all those points x such that $x^{k_l} \rightharpoonup x$ as $l \to \infty$ for some subsequence $\{x^{k_l}\}$ of $\{x^k\}$.

Definition 1 (see [1]) Let $\mathcal{C}$ be a nonempty closed convex subset of real Hilbert space $\mathcal{H}$. An operator $T : \mathcal{C} \to \mathcal{H}$ is called

(1) nonexpansive if $\|Tx - Ty\| \le \|x - y\|$ for all $x, y \in \mathcal{C}$;
(2) firmly nonexpansive if $\|Tx - Ty\|^2 \le \|x - y\|^2 - \|(I - T)x - (I - T)y\|^2$ for all $x, y \in \mathcal{C}$;
(3) η-inverse strongly monotone if exists $\eta > 0$ such that $\langle Tx - Ty, x - y \rangle \ge \eta \|Tx - Ty\|^2$ for all $x, y \in \mathcal{C}$.

For any $x \in \mathcal{H}$, the projection onto a nonempty closed convex subset $\mathcal{C}$ is defined as

$$P_{\mathcal{C}} x = \arg\min\{\|u - x\| \mid u \in \mathcal{C}\}.$$

The projection $P_{\mathcal{C}}$ has the following well-known properties.

Lemma 1 *(see [1]) Let $\mathcal{C} \subseteq \mathcal{H}$ be a nonempty closed convex subset. Then for all $x \in \mathcal{H}$ and $u \in \mathcal{C}$,*

(1) $\langle x - P_{\mathcal{C}} x, u - P_{\mathcal{C}} x \rangle \le 0$;
(2) *$P_{\mathcal{C}}$ and $I - P_{\mathcal{C}}$ are both nonexpansive;*
(3) *$P_{\mathcal{C}}$ and $I - P_{\mathcal{C}}$ are both firmly nonexpansive;*
(4) *$P_{\mathcal{C}}$ and $I - P_{\mathcal{C}}$ are both 1-inverse strongly monotone.*

Definition 2 Let $\lambda \in (0, 1)$ and $f : \mathcal{H} \to \mathbb{R}$ be a function.

(1) f is convex if $f[\lambda x + (1 - \lambda)y] \le \lambda f(x) + (1 - \lambda) f(y),\ \forall x, y \in \mathcal{H}$.
(2) A vector $\xi \in \mathcal{H}$ is a subgradient of f at a point x if

$$f(y) \ge f(x) + \langle \xi, y - x \rangle \quad \forall y \in \mathcal{H}. \tag{17}$$

The set of all subgradients of f at x, denoted by $\partial f(x)$, is called the subdifferential of f.

(3) f is said to be weakly lower semi-continuous (w-lsc) at a point x if $x^k \rightharpoonup x$ implies

$$f(x) \le \liminf_{k \to \infty} f(x^k).$$

Lemma 2 *([12], Lemma 8) Assume $\{s_k\}$ is a sequence of nonnegative real numbers such that*

$$s_{k+1} \le (1 - a_k)s_k + a_k b_k,\ s_{k+1} \le s_k - c_k + d_k, \quad k \ge 1,$$

where $\{a_k\}$ is a sequence in $(0, 1)$, $\{c_k\}$ is a sequence of nonnegative real numbers, and $\{b_k\}$ and $\{d_k\}$ are two sequences in $\mathbb{R}$ such that

(1) $\sum_{k=0}^{\infty} a_k = \infty$;
(2) $\lim_{k\to\infty} d_k = 0$;
(3) $\lim_{l\to\infty} c_{k_l} = 0$ *implies* $\limsup_{l\to\infty} b_{k_l} \le 0$ *for any subsequence* $\{k_l\} \subset \{k\}$.

Then $\lim_{k\to\infty} s_k = 0$.

3 Algorithms and Strong Convergences

In what follows, we will study the SFPMOS (12), under the following assumptions.

(A1) The solution set $\Omega = \{x \in \mathcal{C} \mid \mathcal{F}_j x \in \mathcal{Q}_j,\ \forall j = 1, \ldots, N\}$ is nonempty.
(A2) The sets $\mathcal{C}$ and $\mathcal{Q}_j$, $j = 1, \ldots, N$, are given by

$$\mathcal{C} = \{x \in \mathcal{H} \mid g(x) \le 0\} \text{ and } \mathcal{Q}_j = \{y \in \mathcal{H}_j \mid h_j(y) \le 0\},$$

where $g : \mathcal{H} \to \mathbb{R}$ and $h_j : \mathcal{H}_j \to \mathbb{R}$ are convex, subdifferentiable, and w-lsc functions on $\mathcal{H}$ and $\mathcal{H}_j$, respectively.
(A3) For any $x \in \mathcal{H}$ and $y_j \in \mathcal{H}_j$, at least one subgradient $\xi \in \partial g(x)$ and $\eta_j \in \partial h_j(y_j)$ can be calculated, respectively. We assume also that the subdifferential operators ∂g and ∂h_j are bounded on bounded sets.

Remark 1 It is worth noting that every convex function defined on a finite-dimensional Hilbert space automatically satisfies assumption (A3) (see [2], Corollary 7.9).

Define the sets $\mathcal{C}_k$ and $\mathcal{Q}_{j,k}$, $j = 1, \ldots, N$, at point x^k by

$$\mathcal{C}_k = \{x \in \mathcal{H} \mid g(x^k) + \langle \xi_k, x - x^k\rangle \le 0\},\ \ \xi_k \in \partial g(x^k)$$
$$\mathcal{Q}_{j,k} = \{y \in \mathcal{H}_j \mid h_j(\mathcal{F}_j x^k) + \langle \eta_{j,k}, y - \mathcal{F}_j x^k\rangle \le 0\},\ \ \eta_{j,k} \in \partial h_j(\mathcal{F}_j x^k).$$

For each $x \in \mathcal{H}$, $j = 1, \ldots, N$, and for all $k \ge 1$, define

$$T_{j,k}(x) = \mathcal{F}_j^*(I - P_{\mathcal{Q}_{j,k}})\mathcal{F}_j(x) \text{ and } S_{j,k}(x) = (I - P_{\mathcal{Q}_{j,k}})\mathcal{F}_j(x).$$

3.1 The Relaxed CQ Algorithm

Remark 2 (1) Note that $\mathcal{C}_k$ and $\mathcal{Q}_{j,k}$ are half-spaces and thus the corresponding projections are easily calculated.
(2) It is clear that $\mathcal{C} \subseteq \mathcal{C}_k$ and $\mathcal{Q}_j \subseteq \mathcal{Q}_{j,k}$, $j = 1, \ldots, N$, for all $k \ge 1$ from the subdifferentiable inequality (17).

First, we demonstrate the validity of the stopping criterion of Algorithm 1.

Lemma 3 *If* $x^{k+1} = x^k$ *and* $\|T_{j,k}(x^k)\| = 0$ *for all* $j = 1, \ldots, N$, *for some* k, *then* x^k *is a solution of the* SFPMOS *(12).*

Algorithm 1 The relaxed CQ algorithm

• **Initial Step:** Select two sequences $\{\alpha_k\}$ and $\{\rho_k\}$ satisfying conditions (4) and (9), $\mu \in (0, 1)$. Let $x^0, x^1 \in \mathcal{H}$ be arbitrary. Input $k := 1$.
• **Iterative Step:** Given the iterate x^k $(k \geq 1)$, calculate x^{k+1} as follows.

Step 1 Compute $y^k = x^k - \gamma_k\big[\mu \sum_{j=1}^N T_{j,k}(x^k) + (1-\mu)\sum_{j=1}^N T_{j,k-1}(x^k)\big]$, where the step size γ_k is chosen as

$$\gamma_k = \rho_k \frac{\mu \sum_{j=1}^N \|S_{j,k}(x^k)\|^2 + (1-\mu)\sum_{j=1}^N \|S_{j,k-1}(x^k)\|^2}{\mu \|\sum_{j=1}^N T_{j,k}(x^k)\|^2 + (1-\mu)\|\sum_{j=1}^N T_{j,k-1}(x^k)\|^2}. \tag{18}$$

Step 2 Compute $z^k = (1-\alpha_k)y^k$ and $x^{k+1} = P_{\mathcal{C}_k} P_{\mathcal{C}_{k-1}} z^k$.

• **Stopping Criterion:** If $x^{k+1} = x^k$ and $\|T_{j,k}(x^k)\| = 0$ for all $j = 1, \ldots, N$, then stop. Otherwise, set $k := k+1$ and go to **Iterative Step** to compute the next iterate x^{k+2}.

Proof Let $x^{k+1} = x^k$. It follows from Steps 1 and 2 in Algorithm 1 that

$$x^k = P_{\mathcal{C}_k} P_{\mathcal{C}_{k-1}}\Big[(1-\alpha_k)\Big(x^k - \gamma_k\Big(\mu \sum_{j=1}^N T_{j,k}(x^k) + (1-\mu)\sum_{j=1}^N T_{j,k-1}(x^k)\Big)\Big)\Big],$$

which means that $x^k \in \mathcal{C}_k$. By the definition of $\mathcal{C}_k$, we have $x^k \in \mathcal{C}$. Using $\|T_{j,k}(x^k)\| = 0$ for all $j = 1, \ldots, N$, we conclude that $\mathcal{F}_j(x^k) \in \mathcal{Q}_{j,k}$ for all $j = 1, \ldots, N$. By the definition of $\mathcal{Q}_{j,k}$, we get $\mathcal{F}_j(x^k) \in \mathcal{Q}_j$ for all $j = 1, \ldots, N$. Therefore, $x^k \in \Omega$. $\square$

Lemma 4 *Let $\{y^k\}$ and $\{z^k\}$ be two sequences generated by Algorithm 1. Then, for all $k \geq 1$ and $u \in \Omega$, we have*

$$\|y^k - u\|^2 \leq \|x^k - u\|^2 - \rho_k(2-\rho_k)\frac{\big[\mu \sum_{j=1}^N \|S_{j,k}(x^k)\|^2 + (1-\mu)\sum_{j=1}^N \|S_{j,k-1}(x^k)\|^2\big]^2}{\mu \|\sum_{j=1}^N T_{j,k}(x^k)\|^2 + (1-\mu)\|\sum_{j=1}^N T_{j,k-1}(x^k)\|^2} \tag{19}$$

and

$$\|z^k - u\|^2 \leq \alpha_k \|u\|^2 + (1-\alpha_k)\|x^k - u\|^2 - \rho_k(2-\rho_k)(1-\alpha_k)\frac{\big[\mu \sum_{j=1}^N \|S_{j,k}(x^k)\|^2 + (1-\mu)\sum_{j=1}^N \|S_{j,k-1}(x^k)\|^2\big]^2}{\mu \|\sum_{j=1}^N T_{j,k}(x^k)\|^2 + (1-\mu)\|\sum_{j=1}^N T_{j,k-1}(x^k)\|^2}. \tag{20}$$

Proof Let $u \in \Omega$. Hence, $\mathcal{F}_j u \in \mathcal{Q}_j$ for each $j = 1, \ldots, N$. It follows from $\mathcal{Q}_j \subseteq \mathcal{Q}_{j,k-1}$ and $\mathcal{Q}_j \subseteq \mathcal{Q}_{j,k}$ for all $k \geq 1$ that

$$\mathcal{F}_j u = P_{\mathcal{Q}_{j,k}} \mathcal{F}_j u = P_{\mathcal{Q}_{j,k-1}} \mathcal{F}_j u. \tag{21}$$

From Step 1 in Algorithm 1, the property of adjoint operator $\mathcal{F}_j^*$, the 1-inverse strong monotonicity of $I - P_{Q_{j,k}}$ and $I - P_{Q_{j,k-1}}$, the convexity of $\|.\|^2$, and (21), we get

$$\begin{aligned}
\|y^k - u\|^2 &= \|x^k - u\|^2 + \gamma_k^2 \Big\| \mu \sum_{j=1}^N T_{j,k}(x^k) + (1-\mu) \sum_{j=1}^N T_{j,k-1}(x^k) \Big\|^2 \\
&\quad - 2\gamma_k \Big\langle \mu \sum_{j=1}^N T_{j,k}(x^k) + (1-\mu) \sum_{j=1}^N T_{j,k-1}(x^k), x^k - u \Big\rangle \\
&= \|x^k - u\|^2 + \gamma_k^2 \Big\| \mu \sum_{j=1}^N T_{j,k}(x^k) + (1-\mu) \sum_{j=1}^N T_{j,k-1}(x^k) \Big\|^2 \\
&\quad - 2\gamma_k \mu \sum_{j=1}^N \big\langle S_{j,k}(x^k) - S_{j,k}(u), \mathcal{F}_j x^k - \mathcal{F}_j u \big\rangle \\
&\quad - 2\gamma_k (1-\mu) \sum_{j=1}^N \big\langle S_{j,k-1}(x^k) - S_{j,k-1}(u), \mathcal{F}_j x^k - \mathcal{F}_j u \big\rangle \\
&\le \|x^k - u\|^2 + \gamma_k^2 \Big[\mu \Big\| \sum_{j=1}^N T_{j,k}(x^k) \Big\|^2 + (1-\mu) \Big\| \sum_{j=1}^N T_{j,k-1}(x^k) \Big\|^2 \Big] \\
&\quad - 2\gamma_k \Big[\mu \sum_{j=1}^N \big\| S_{j,k}(x^k) \big\|^2 + (1-\mu) \sum_{j=1}^N \big\| S_{j,k-1}(x^k) \big\|^2 \Big], \quad k \ge 1.
\end{aligned} \tag{22}$$

Substituting (18) into (22), we get, for all $k \ge 1$,

$$\begin{aligned}
\|y^k - u\|^2 &\le \|x^k - u\|^2 + \left\{ \rho_k \frac{\mu \sum_{j=1}^N \|S_{j,k}(x^k)\|^2 + (1-\mu) \sum_{j=1}^N \|S_{j,k-1}(x^k)\|^2}{\mu \| \sum_{j=1}^N T_{j,k}(x^k)\|^2 + (1-\mu) \| \sum_{j=1}^N T_{j,k-1}(x^k)\|^2} \right\}^2 \\
&\quad \times \Big[\mu \Big\| \sum_{j=1}^N T_{j,k}(x^k) \Big\|^2 + (1-\mu) \Big\| \sum_{j=1}^N T_{j,k-1}(x^k) \Big\|^2 \Big] \\
&\quad - 2\rho_k \frac{\mu \sum_{j=1}^N \|S_{j,k}(x^k)\|^2 + (1-\mu) \sum_{j=1}^N \|S_{j,k-1}(x^k)\|^2}{\mu \| \sum_{j=1}^N T_{j,k}(x^k)\|^2 + (1-\mu) \| \sum_{j=1}^N T_{j,k-1}(x^k)\|^2} \\
&\quad \times \Big[\mu \sum_{j=1}^N \big\| S_{j,k}(x^k) \big\|^2 + (1-\mu) \sum_{j=1}^N \big\| S_{j,k-1}(x^k) \big\|^2 \Big] \\
&= \|x^k - u\|^2 - \rho_k (2 - \rho_k) \frac{\big[\mu \sum_{j=1}^N \|S_{j,k}(x^k)\|^2 + (1-\mu) \sum_{j=1}^N \|S_{j,k-1}(x^k)\|^2\big]^2}{\mu \| \sum_{j=1}^N T_{j,k}(x^k)\|^2 + (1-\mu) \| \sum_{j=1}^N T_{j,k-1}(x^k)\|^2}.
\end{aligned}$$

Thus, (19) holds.

Now, it follows from Steps 1, 2 in Algorithm 1 and the convexity of $\|.\|^2$ that

$$\|z^k - u\|^2 = \|(1-\alpha_k)y^k - u\|^2 = \|\alpha_k(-u) + (1-\alpha_k)(y^k - u)\|^2$$
$$\leq \alpha_k\|u\|^2 + (1-\alpha_k)\|y^k - u\|^2, \quad k \geq 1. \tag{23}$$

From (19) and (23) we get (20). This completes the proof of the lemma. □

Proposition 5 *Let $\{x^k\}$ be a sequence generated by Algorithm 1. Then*

(i) The sequence $\{x^k\}$ is bounded.
(ii) For any $u \in \Omega$, for all $k \geq 1$, the following inequalities hold:

$$\|x^{k+1} - u\|^2 \leq (1-\alpha_k)\|x^k - u\|^2 + \alpha_k b_k, \tag{24}$$
$$\|x^{k+1} - u\|^2 \leq \|x^k - u\|^2 - c_k + \alpha_k\|u\|^2, \tag{25}$$

where

$$b_k = \alpha_k\|u\|^2 + 2(1-\alpha_k)\langle x^k - u, -u\rangle$$
$$+ 2(1-\alpha_k)\gamma_k\Big[\mu\Big\|\sum_{j=1}^N T_{j,k}(x^k)\Big\| + (1-\mu)\Big\|\sum_{j=1}^N T_{j,k-1}(x^k)\Big\|\Big]\|u\|,$$
$$c_k = \varepsilon\frac{\big[\mu\sum_{j=1}^N\|S_{j,k}(x^k)\|^2 + (1-\mu)\sum_{j=1}^N\|S_{j,k-1}(x^k)\|^2\big]^2}{\mu\|\sum_{j=1}^N T_{j,k}(x^k)\|^2 + (1-\mu)\|\sum_{j=1}^N T_{j,k-1}(x^k)\|^2}$$
$$+ \|(I - P_{\mathcal{C}_{k-1}})z^k\|^2 + \|(I - P_{\mathcal{C}_k})P_{\mathcal{C}_{k-1}}z^k\|^2, \ \varepsilon > 0.$$

Proof (i) Since $u \in \Omega$, $u \in \mathcal{C}$. From $\mathcal{C} \subseteq \mathcal{C}_{k-1}$ and $\mathcal{C} \subseteq \mathcal{C}_k$, we get

$$u = P_{\mathcal{C}_k}u = P_{\mathcal{C}_{k-1}}u. \tag{26}$$

Since $P_{\mathcal{C}_k}$ and $P_{\mathcal{C}_{k-1}}$ are both firmly nonexpansive, from Step 2 in Algorithm 1, (20), and (26), we get

$$\|x^{k+1} - u\|^2 = \|P_{\mathcal{C}_k}P_{\mathcal{C}_{k-1}}z^k - P_{\mathcal{C}_k}P_{\mathcal{C}_{k-1}}u\|^2$$
$$\leq \|P_{\mathcal{C}_{k-1}}z^k - P_{\mathcal{C}_{k-1}}u\|^2 - \|(I - P_{\mathcal{C}_k})P_{\mathcal{C}_{k-1}}z^k - (I - P_{\mathcal{C}_k})P_{\mathcal{C}_{k-1}}u\|^2$$
$$\leq \|z^k - u\|^2 - \|(I - P_{\mathcal{C}_{k-1}})z^k\|^2 - \|(I - P_{\mathcal{C}_k})P_{\mathcal{C}_{k-1}}z^k\|^2$$
$$\leq \alpha_k\|u\|^2 + (1-\alpha_k)\|x^k - u\|^2$$
$$- \rho_k(2-\rho_k)(1-\alpha_k)\frac{\big[\mu\sum_{j=1}^N\|S_{j,k}(x^k)\|^2 + (1-\mu)\sum_{j=1}^N\|S_{j,k-1}(x^k)\|^2\big]^2}{\mu\|\sum_{j=1}^N T_{j,k}(x^k)\|^2 + (1-\mu)\|\sum_{j=1}^N T_{j,k-1}(x^k)\|^2}$$
$$- \|(I - P_{\mathcal{C}_{k-1}})z^k\|^2 - \|(I - P_{\mathcal{C}_k})P_{\mathcal{C}_{k-1}}z^k\|^2. \tag{27}$$

Using (27), (4), and (9), we have

$$\|x^{k+1} - u\|^2 \leq \alpha_k\|u\|^2 + (1-\alpha_k)\|x^k - u\|^2$$

$$
\begin{aligned}
&\le \max\{\|u\|^2, \|x^k - u\|^2\}\\
&\le \dots\\
&\le \max\{\|u\|^2, \|x^0 - u\|^2\}.
\end{aligned}
$$

Therefore, the sequence $\{\|x^k - u\|\}$ is bounded. Consequently, the sequence $\{x^k\}$ is bounded.

(ii) It follows from (19) and (9) that, for all $k \ge 1$,

$$\|y^k - u\|^2 \le \|x^k - u\|^2. \tag{28}$$

Since $P_{\mathcal{C}_{k-1}}$ and $P_{\mathcal{C}_k}$ are both nonexpansive, from Steps 1, 2 in Algorithm 1, and (28), we obtain

$$
\begin{aligned}
\|x^{k+1} - u\|^2 &= \|P_{\mathcal{C}_k} P_{\mathcal{C}_{k-1}} z^k - P_{\mathcal{C}_k} P_{\mathcal{C}_{k-1}} u\|^2 \le \|z^k - u\|^2 = \|(1-\alpha_k) y^k - u\|^2\\
&= \|\alpha_k(-u) + (1-\alpha_k)(y^k - u)\|^2\\
&= \alpha_k^2 \|u\|^2 + (1-\alpha_k)^2 \|y^k - u\|^2 + 2\alpha_k(1-\alpha_k)\langle y^k - u, -u\rangle\\
&\le \alpha_k^2 \|u\|^2 + (1-\alpha_k)^2 \|x^k - u\|^2 + 2\alpha_k(1-\alpha_k)\langle x^k - u, -u\rangle\\
&\quad + 2\alpha_k(1-\alpha_k)\gamma_k\Big[\mu\Big\|\sum_{j=1}^N T_{j,k}(x^k)\Big\| + (1-\mu)\Big\|\sum_{j=1}^N T_{j,k-1}(x^k)\Big\|\Big]\|u\|\\
&\le (1-\alpha_k)\|x^k - u\|^2 + \alpha_k\Big[\alpha_k\|u\|^2 + 2(1-\alpha_k)\langle x^k - u, -u\rangle\\
&\quad + 2(1-\alpha_k)\gamma_k\Big(\mu\Big\|\sum_{j=1}^N T_{j,k}(x^k)\Big\| + (1-\mu)\Big\|\sum_{j=1}^N T_{j,k-1}(x^k)\Big\|\Big)\|u\|\Big]\\
&= (1-\alpha_k)\|x^k - u\|^2 + \alpha_k b_k.
\end{aligned}
$$

Thus, (24) holds.

According to (4) and (9), we may assume that there exists $\varepsilon > 0$ such that $\rho_k(2 - \rho_k)(1-\alpha_k) \ge \varepsilon$. Furthermore, we get using (27) that

$$
\begin{aligned}
\|x^{k+1} - u\|^2 &\le \|x^k - u\|^2 - \varepsilon \frac{\big[\mu \sum_{j=1}^N \|S_{j,k}(x^k)\|^2 + (1-\mu)\sum_{j=1}^N \|S_{j,k-1}(x^k)\|^2\big]^2}{\mu\|\sum_{j=1}^N T_{j,k}(x^k)\|^2 + (1-\mu)\|\sum_{j=1}^N T_{j,k-1}(x^k)\|^2}\\
&\quad - \|(I - P_{\mathcal{C}_{k-1}}) z^k\|^2 - \|(I - P_{\mathcal{C}_k}) P_{\mathcal{C}_{k-1}} z^k\|^2 + \alpha_k \|u\|^2\\
&= \|x^k - u\|^2 - c_k + \alpha_k\|u\|^2, \quad k \ge 1.
\end{aligned}
$$

Thus, (25) holds. □

By Lemma 3, we may assume that the sequence $\{x^k\}$ generated by Algorithm 1 is infinite. In other words, Algorithm 1 does not terminate in a finite number of iterations. Its convergence is proved below.

Theorem 6 *Let $\{x^k\}$ be a sequence generated by Algorithm 1. Then the sequence $\{x^k\}$ converges strongly to an element $u^* \in \Omega$, where $u^* = P_\Omega(0)$.*

Proof Since the solution set Ω of (12) is nonempty, closed and convex, there exists unique $u^* = P_\Omega(0)$. In particular, $u^* \in \Omega$. Now, it follows from Proposition 5(ii) with u replaced by u^* that

$$\|x^{k+1} - u^*\|^2 \leq (1 - \alpha_k)\|x^k - u^*\|^2 + \alpha_k b_k^*,$$
$$\|x^{k+1} - u^*\|^2 \leq \|x^k - u^*\|^2 - c_k^* + \alpha_k \|u^*\|^2,$$

where

$$b_k^* = \alpha_k \|u^*\|^2 + 2(1 - \alpha_k)\langle x^k - u^*, -u^*\rangle + 2(1 - \alpha_k)\gamma_k \Big[\mu \Big\| \sum_{j=1}^{N} T_{j,k}(x^k)\Big\|$$
$$+ (1 - \mu)\Big\| \sum_{j=1}^{N} T_{j,k-1}(x^k)\Big\|\Big]\|u^*\|,$$
$$c_k^* = \varepsilon \frac{\left[\mu \sum_{j=1}^{N} \|S_{j,k}(x^k)\|^2 + (1 - \mu) \sum_{j=1}^{N} \|S_{j,k-1}(x^k)\|^2\right]^2}{\mu \| \sum_{j=1}^{N} T_{j,k}(x^k)\|^2 + (1 - \mu)\| \sum_{j=1}^{N} T_{j,k-1}(x^k)\|^2} + \|(I - P_{\mathcal{C}_{k-1}})z^k\|^2$$
$$+ \|(I - P_{\mathcal{C}_k})P_{\mathcal{C}_{k-1}} z^k\|^2.$$

From (4), $\alpha_k \in (0, 1)$, $\sum_{k=1}^{\infty} \alpha_k = \infty$ and $\alpha_k \|u^*\| \to 0$ as $k \to \infty$. Let the data $s_k = \|x^k - u^*\|^2$, to use Lemma 2, it suffices to verify that for every subsequence $\{k_l\}$ of $\{k\}$, $\lim_{l\to\infty} c_{k_l}^* = 0$ implies $\limsup_{l\to\infty} b_{k_l}^* \leq 0$. Indeed, suppose that $\lim_{l\to\infty} c_{k_l}^* = 0$, then we obtain

$$\lim_{l\to\infty} \Bigg\{ \varepsilon \frac{\left[\mu \sum_{j=1}^{N} \|S_{j,k_l}(x^{k_l})\|^2 + (1 - \mu) \sum_{j=1}^{N} \|S_{j,k_l-1}(x^{k_l})\|^2\right]^2}{\mu \| \sum_{j=1}^{N} T_{j,k_l}(x^{k_l})\|^2 + (1 - \mu)\| \sum_{j=1}^{N} T_{j,k_l-1}(x^{k_l})\|^2}$$
$$+ \|(I - P_{\mathcal{C}_{k_l-1}})z^{k_l}\|^2 + \|(I - P_{\mathcal{C}_{k_l}})P_{\mathcal{C}_{k_l-1}} z^{k_l}\|^2 \Bigg\} = 0. \tag{29}$$

This implies that

$$\lim_{l\to\infty} \frac{\left[\mu \sum_{j=1}^{N} \|S_{j,k_l}(x^{k_l})\|^2 + (1 - \mu) \sum_{j=1}^{N} \|S_{j,k_l-1}(x^{k_l})\|^2\right]^2}{\mu \| \sum_{j=1}^{N} T_{j,k_l}(x^{k_l})\|^2 + (1 - \mu)\| \sum_{j=1}^{N} T_{j,k_l-1}(x^{k_l})\|^2} = 0, \tag{30}$$

and

$$\lim_{l\to\infty} \|(I - P_{\mathcal{C}_{k_l-1}})z^{k_l}\|^2 = 0, \tag{31}$$

and

$$\lim_{l\to\infty} \|(I - P_{\mathcal{C}_{k_l}})P_{\mathcal{C}_{k_l-1}} z^{k_l}\|^2 = 0. \tag{32}$$

Since $I - P_{\mathcal{Q}_{j,k_l}}$ and $I - P_{\mathcal{Q}_{j,k_l-1}}$ are both nonexpansive, $\mathcal{F}_j$ is a bounded linear operator, and (21), we have

$$\begin{aligned}
&\mu \Big\| \sum_{j=1}^N T_{j,k_l}(x^{k_l}) \Big\|^2 + (1-\mu) \Big\| \sum_{j=1}^N T_{j,k_l-1}(x^{k_l}) \Big\|^2 \\
&= \mu \Big\| \sum_{j=1}^N \Big(T_{j,k_l}(x^{k_l}) - T_{j,k_l}(u^*) \Big) \Big\|^2 \\
&\quad + (1-\mu) \Big\| \sum_{j=1}^N \Big(T_{j,k_l-1}(x^{k_l}) - T_{j,k_l-1}(u^*) \Big) \Big\|^2 \\
&\leq \mu \sum_{j=1}^N \|\mathcal{F}_j\|^4 \|x^{k_l} - u^*\|^2 + (1-\mu) \sum_{j=1}^N \|\mathcal{F}_j\|^4 \|x^{k_l} - u^*\|^2 \\
&\leq NM \|x^{k_l} - u^*\|^2, \; M = \max_{j=1,\ldots,N} \{\|\mathcal{F}_j\|^4\}.
\end{aligned} \tag{33}$$

Since $\{x^{k_l}\}$ is bounded, $\mathcal{F}_j$ is a bounded linear operator, it follows that $\big\{\mu \big\| \sum_{j=1}^N T_{j,k_l}(x^{k_l}) \big\|^2 + (1-\mu) \big\| \sum_{j=1}^N T_{j,k_l-1}(x^{k_l}) \big\|^2 \big\}$ is bounded, which, together with (30) implies that

$$\lim_{l\to\infty} \Big\{ \mu \sum_{j=1}^N \|S_{j,k_l}(x^{k_l})\|^2 + (1-\mu) \sum_{j=1}^N \|S_{j,k_l-1}(x^{k_l})\|^2 \Big\} = 0. \tag{34}$$

In particular, we have

$$\lim_{l\to\infty} \|S_{j,k_l}(x^{k_l})\|^2 = 0 \text{ for all } j = 1, \ldots, N. \tag{35}$$

Now, we show that $\limsup_{l\to\infty} b^*_{k_l} \leq 0$. Indeed, take a subsequence $\{x^{k_{l_m}}\}$ of $\{x^{k_l}\}$ such that

$$\limsup_{l\to\infty} \langle x^{k_l} - u^*, -u^* \rangle = \lim_{m\to\infty} \langle x^{k_{l_m}} - u^*, -u^* \rangle.$$

Since the sequence $\{x^{k_{l_m}}\}$ is bounded, there exists a subsequence of $\{x^{k_{l_m}}\}$ which converges weakly to $\widehat{u}$. We may assume, without any loss of generality, that $x^{k_{l_m}} \rightharpoonup \widehat{u}$. Since $P_{\mathcal{Q}_{j,k_{l_m}}} \mathcal{F}_j x^{k_{l_m}} \in \mathcal{Q}_{j,k_{l_m}}$ for all $j = 1, \ldots, N$, we have

$$h_j\big(\mathcal{F}_j x^{k_{l_m}}\big) \leq \langle \beta_{j,k_{l_m}}, S_{j,k_{l_m}}(x^{k_{l_m}}) \rangle,$$

where $\beta_{j,k_{l_m}} \in \partial h_j(\mathcal{F}_j x^{k_{l_m}})$. By the hypothesis that $\{\beta_{j,k_{l_m}}\}$ is bounded and (35), one has

$$h_j\big(\mathcal{F}_j x^{k_{l_m}}\big) \leq \|\beta_{j,k_{l_m}}\| \|S_{j,k_{l_m}}(x^{k_{l_m}})\| \to 0 \text{ as } m \to \infty \text{ for all } j = 1, \dots, N. \tag{36}$$

Moreover, since each $\mathcal{F}_j$ is a bounded linear operator and $x^{k_{l_m}} \rightharpoonup \widehat{u}$, $\mathcal{F}_j x^{k_{l_m}} \rightharpoonup \mathcal{F}_j \widehat{u}$. Since h_j is w-lsc, it follows from (36) that

$$h_j\big(\mathcal{F}_j \widehat{u}\big) \leq \liminf_{m\to\infty} h_j\big(\mathcal{F}_j x^{k_{l_m}}\big) \leq 0, \quad \text{for all } j = 1, \dots, N,$$

which means that $\mathcal{F}_j \widehat{u} \in \mathcal{Q}_j$ for all $j = 1, \dots, N$.

On the other hand, combining Steps 1 and 2 in Algorithm 1 with (30), we have

$$\begin{aligned}
&\|z^{k_{l_m}} - x^{k_{l_m}}\|^2 \\
&= \Big\|(1-\alpha_{k_{l_m}})\Big[x^{k_{l_m}} - \gamma_{k_{l_m}}\Big(\mu \sum_{j=1}^{N} T_{j,k_{l_m}}(x^{k_{l_m}}) + (1-\mu)\sum_{j=1}^{N} T_{j,k_{l_m}-1}(x^{k_{l_m}})\Big)\Big] \\
&- x^{k_{l_m}}\Big\|^2 \\
&= \Big\|\alpha_{k_{l_m}}(-x^{k_{l_m}}) + (1-\alpha_{k_{l_m}})\Big[x^{k_{l_m}} - \gamma_{k_{l_m}}\Big(\mu \sum_{j=1}^{N} T_{j,k_{l_m}}(x^{k_{l_m}}) \\
&+ (1-\mu)\sum_{j=1}^{N} T_{j,k_{l_m}-1}(x^{k_{l_m}})\Big)\Big] - x^{k_{l_m}}\Big\|^2 \\
&\leq \alpha_{k_{l_m}}\|x^{k_{l_m}}\|^2 + (1-\alpha_{k_{l_m}})\gamma_{k_{l_m}}^2 \Big\|\mu \sum_{j=1}^{N} T_{j,k_{l_m}}(x^{k_{l_m}}) + (1-\mu)\sum_{j=1}^{N} T_{j,k_{l_m}-1}(x^{k_{l_m}})\Big\|^2 \\
&\leq \alpha_{k_{l_m}}\|x^{k_{l_m}}\|^2 \\
&+ (1-\alpha_{k_{l_m}})\gamma_{k_{l_m}}^2 \Big[\mu\Big\|\sum_{j=1}^{N} T_{j,k_{l_m}}(x^{k_{l_m}})\Big\|^2 + (1-\mu)\Big\|\sum_{j=1}^{N} T_{j,k_{l_m}-1}(x^{k_{l_m}})\Big\|^2\Big] \\
&= (1-\alpha_{k_{l_m}})\rho_{k_{l_m}}^2 \frac{\big[\mu \sum_{j=1}^{N} \|S_{j,k_{l_m}}(x^{k_{l_m}})\|^2 + (1-\mu)\sum_{j=1}^{N} \|S_{j,k_{l_m}-1}(x^{k_{l_m}})\|^2\big]^2}{\mu\|\sum_{j=1}^{N} T_{j,k_{l_m}}(x^{k_{l_m}})\|^2 + (1-\mu)\|\sum_{j=1}^{N} T_{j,k_{l_m}-1}(x^{k_{l_m}})\|^2} \\
&+ \alpha_{k_{l_m}}\|x^{k_{l_m}}\|^2 \to 0.
\end{aligned} \tag{37}$$

It follows from (31) and (32) that

$$\begin{aligned}
\|x^{k_{l_m}+1} - z^{k_{l_m}}\|^2 &= \Big\|P_{\mathcal{C}_{k_{l_m}}} P_{\mathcal{C}_{k_{l_m}-1}} z^{k_{l_m}} - z^{k_{l_m}}\Big\|^2 \\
&= \Big\|P_{\mathcal{C}_{k_{l_m}}} P_{\mathcal{C}_{k_{l_m}-1}} z^{k_{l_m}} + P_{\mathcal{C}_{k_{l_m}-1}} z^{k_{l_m}} - P_{\mathcal{C}_{k_{l_m}-1}} z^{k_{l_m}} - z^{k_{l_m}}\Big\|^2
\end{aligned}$$

$$\leq 2\left\|(I - P_{C_{k_{l_m}}})P_{C_{k_{l_m}-1}} z^{k_{l_m}}\right\|^2 + 2\left\|(I - P_{C_{k_{l_m}-1}})z^{k_{l_m}}\right\|^2$$
$$\to 0 \text{ as } m \to \infty. \tag{38}$$

In fact, since $x^{k_{l_m}+1} \in C_{k_{l_m}}$ it follows from the definition of $C_{k_{l_m}}$ that

$$\begin{aligned} g(x^{k_{l_m}}) &\leq \langle \xi_{k_{l_m}}, x^{k_{l_m}} - x^{k_{l_m}+1}\rangle = \langle \xi_{k_{l_m}}, (x^{k_{l_m}} - z^{k_{l_m}}) + (z^{k_{l_m}} - x^{k_{l_m}+1})\rangle \\ &= \langle \xi_{k_{l_m}}, x^{k_{l_m}} - z^{k_{l_m}}\rangle + \langle \xi_{k_{l_m}}, z^{k_{l_m}} - x^{k_{l_m}+1}\rangle, \end{aligned} \tag{39}$$

where $\xi_{k_{l_m}} \in \partial g(x^{k_{l_m}})$. By the hypothesis that $\{\xi_{k_{l_m}}\}$ is bounded, (37) and (38), we obtain

$$g(x^{k_{l_m}}) \leq \|\xi_{k_{l_m}}\|\|x^{k_{l_m}} - z^{k_{l_m}}\| + \|\xi_{k_{l_m}}\|\|x^{k_{l_m}+1} - z^{k_{l_m}}\| \to 0 \text{ as } m \to \infty. \tag{40}$$

Since $x^{k_{l_m}} \rightharpoonup \widehat{u}$ and g is w-lsc, it follows from (40) that

$$g(\widehat{u}) \leq \liminf_{m\to\infty} g(x^{k_{l_m}}) \leq 0, \tag{41}$$

which means that $\widehat{u} \in C$. Hence, we conclude that $\widehat{u} \in \Omega$.

From Lemma 1(1) and (4), we now deduce that,

$$\begin{aligned} \limsup_{l\to\infty} b^*_{k_l} &= \limsup_{l\to\infty} \Bigg\{\alpha_{k_l}\|u^*\|^2 + 2(1-\alpha_{k_l})\langle x^{k_l} - u^*, -u^*\rangle \\ &\quad + 2(1-\alpha_{k_l})\gamma_{k_l}\Bigg[\mu\Bigg\|\sum_{j=1}^N T_{j,k_l}(x^{k_l})\Bigg\| \\ &\quad + (1-\mu)\Bigg\|\sum_{j=1}^N T_{j,k_l-1}(x^{k_l})\Bigg\|\Bigg]\|u^*\|\Bigg\} \\ &= 2\limsup_{l\to\infty}\langle x^{k_l} - u^*, -u^*\rangle \\ &= 2\lim_{m\to\infty}\langle x^{k_{l_m}} - u^*, -u^*\rangle = \langle \widehat{u} - u^*, -u^*\rangle \leq 0. \end{aligned}$$

From Lemma 2, we conclude that $x^k \to u^* = P_\Omega(0)$. The proof is complete. □

3.2 The Inertial Relaxed CQ Algorithm

Remark 3 The inertial calculation criterion (42) is easy to implement since the term $\|x^k - x^{k-1}\|$ is known before calculating θ_k. Moreover, it follows from (42) and (4) that $\lim_{k\to\infty}\frac{\theta_k}{\alpha_k}\|x^k - x^{k-1}\| = 0$. Indeed, we obtain $\theta_k\|x^k - x^{k-1}\| \leq \eta_k$ for all $k \geq 1$ which together with $\lim_{k\to\infty}\frac{\eta_k}{\alpha_k} = 0$ implies that $\lim_{k\to\infty}\frac{\theta_k}{\alpha_k}\|x^k - x^{k-1}\| \leq \lim_{k\to\infty}\frac{\eta_k}{\alpha_k} = 0$.

Algorithm 2 The inertial relaxed CQ algorithm

• **Initial Step:** Select two sequences $\{\alpha_k\}$ and $\{\rho_k\}$ satisfying conditions (4) and (9), a sequence $\{\eta_k\}$ such that $\lim_{k\to\infty} \frac{\eta_k}{\alpha_k} = 0$, and $\mu \in (0, 1)$. Let $x^0, x^1 \in \mathcal{H}$ be arbitrary. Input $k := 1$.

• **Iterative Step:** Given the iterates x^{k-1} and x^k $(k \geq 1)$, calculate x^{k+1} as follows.

Step 1 Compute $w^k = x^k + \theta_k \left(x^k - x^{k-1}\right)$, where

$$\theta_k = \begin{cases} \min\left\{\frac{\eta_k}{\|x^k - x^{k-1}\|}, \theta\right\}, & \text{if } x^k \neq x^{k-1}, \\ \theta, & \text{otherwise.} \end{cases} \tag{42}$$

Step 2 Compute $y^k = w^k - \gamma_k\big[\mu \sum_{j=1}^N T_{j,k}(w^k) + (1-\mu)\sum_{j=1}^N T_{j,k-1}(w^k)\big]$, where the step size γ_k is chosen as

$$\gamma_k = \rho_k \frac{\mu \sum_{j=1}^N \|S_{j,k}(w^k)\|^2 + (1-\mu)\sum_{j=1}^N \|S_{j,k-1}(w^k)\|^2}{\mu\|\sum_{j=1}^N T_{j,k}(w^k)\|^2 + (1-\mu)\|\sum_{j=1}^N T_{j,k-1}(w^k)\|^2}. \tag{$\gamma 2$}$$

Step 3 Compute $z^k = (1-\alpha_k)y^k$ and $x^{k+1} = P_{\mathcal{C}_k} P_{\mathcal{C}_{k-1}} z^k$.

• **Stopping Criterion:** If $x^{k+1} = x^k = x^{k-1}$ and $\|T_{j,k}(w^k)\| = 0$ for all $j = 1, \dots, N$, then stop. Otherwise, set $k := k+1$ and go to **Iterative Step** to compute the next iterate x^{k+2}.

Lemma 7 *If $x^{k+1} = x^k = x^{k-1}$ and $\|T_{j,k}(w^k)\| = 0$ for all $j = 1, \dots, N$, for some k, then x^k is a solution of the (12).*

Proof Let $x^{k+1} = x^k = x^{k-1}$. It follows from Steps 1, 2, and 3 in Algorithm 2 that $w^k = x^k$ and

$$x^k = P_{\mathcal{C}_k} P_{\mathcal{C}_{k-1}}\Big[(1-\alpha_k)\Big(x^k - \gamma_k\big(\mu \sum_{j=1}^N T_{j,k}(x^k) + (1-\mu)\sum_{j=1}^N T_{j,k-1}(x^k)\big)\Big)\Big],$$

which means that $x^k \in \mathcal{C}_k$. By the definition of $\mathcal{C}_k$, we have $x^k \in \mathcal{C}$. Using $\|T_{j,k}(x^k)\| = \|T_{j,k}(w^k)\| = 0$ for all $j = 1, \dots, N$, we conclude that $\mathcal{F}_j(x^k) \in \mathcal{Q}_{j,k}$ for all $j = 1, \dots, N$. By the definition of $\mathcal{Q}_{j,k}$, we get $\mathcal{F}_j(x^k) \in \mathcal{Q}_j$ for all $j = 1, \dots, N$. The proof is complete. □

Lemma 8 *Let $\{y^k\}$ and $\{z^k\}$ be two sequences generated by Algorithm 2. Then, for all $k \geq 1$ and $u \in \Omega$, we get*

$$\|y^k - u\|^2 \leq \|w^k - u\|^2 - \rho_k(2-\rho_k)\frac{\big[\mu \sum_{j=1}^N \|S_{j,k}(w^k)\|^2 + (1-\mu)\sum_{j=1}^N \|S_{j,k-1}(w^k)\|^2\big]^2}{\mu\|\sum_{j=1}^N T_{j,k}(w^k)\|^2 + (1-\mu)\|\sum_{j=1}^N T_{j,k-1}(w^k)\|^2} \tag{43}$$

and

$$\|z^k - u\|^2 \leq \alpha_k\|u\|^2 + (1-\alpha_k)\|w^k - u\|^2$$

$$- \rho_k(2-\rho_k)(1-\alpha_k)\frac{\left[\mu \sum_{j=1}^N \|S_{j,k}(w^k)\|^2 + (1-\mu)\sum_{j=1}^N \|S_{j,k-1}(w^k)\|^2\right]^2}{\mu\|\sum_{j=1}^N T_{j,k}(w^k)\|^2 + (1-\mu)\|\sum_{j=1}^N T_{j,k-1}(w^k)\|^2}. \tag{44}$$

Proof Using the same arguments as in the proof of Lemma 4. □

Proposition 9 *Let $\{x^k\}$ be a sequence generated by Algorithm 2. Then*

(i) The sequence $\{x^k\}$ is bounded.

(ii) For any $u \in \Omega$, for all $k \geq 1$, for some $M_2 > 0$, the following inequalities hold:

$$\|x^{k+1}-u\|^2 \leq (1-\alpha_k)\|x^k-u\|^2 + \alpha_k\widehat{b}_k, \tag{45}$$

$$\|x^{k+1}-u\|^2 \leq \|x^k-u\|^2 - \widehat{c}_k + \alpha_k\|u\|^2 + (1-\alpha_k)\theta_k\|x^k-x^{k-1}\|M_2, \tag{46}$$

where

$$\begin{aligned}\widehat{b}_k &= \alpha_k\|u\|^2 + 2(1-\alpha_k)\langle w^k-u, -u\rangle + (1-\alpha_k)\frac{\theta_k}{\alpha_k}\|x^k-x^{k-1}\|M_2 \\ &\quad + 2(1-\alpha_k)\gamma_k\Big[\mu\Big\|\sum_{j=1}^N T_{j,k}(w^k)\Big\| + (1-\mu)\Big\|\sum_{j=1}^N T_{j,k-1}(w^k)\Big\|\Big]\|u\|, \\ \widehat{c}_k &= \varepsilon\frac{\left[\mu \sum_{j=1}^N \|S_{j,k}(w^k)\|^2 + (1-\mu)\sum_{j=1}^N \|S_{j,k-1}(w^k)\|^2\right]^2}{\mu\|\sum_{j=1}^N T_{j,k}(w^k)\|^2 + (1-\mu)\|\sum_{j=1}^N T_{j,k-1}(w^k)\|^2} \\ &\quad + \|(I-P_{\mathcal{C}_{k-1}})z^k\|^2 + \|(I-P_{\mathcal{C}_k})P_{\mathcal{C}_{k-1}}z^k\|^2.\end{aligned}$$

Proof (i) It follows from (43), (4), and (9) that

$$\|y^k-u\| \leq \|w^k-u\|, \quad k \geq 1. \tag{47}$$

From Step 3 in Algorithm 2, the non-expansiveness of $P_{\mathcal{C}_k}$ and $P_{\mathcal{C}_{k-1}}$, and the convexity of $\|.\|$, we get

$$\begin{aligned}\|x^{k+1}-u\| &= \|P_{\mathcal{C}_k}P_{\mathcal{C}_{k-1}}z^k - P_{\mathcal{C}_k}P_{\mathcal{C}_{k-1}}u\| \leq \|(1-\alpha_k)y^k-u\| \\ &= \|(1-\alpha_k)(y^k-u)+\alpha_k(-u)\| \leq (1-\alpha_k)\|y^k-u\| + \alpha_k\|u\|. \end{aligned} \tag{48}$$

From Step 1 in Algorithm 2, we have

$$\|w^k-u\| = \|x^k+\theta_k\big(x^k-x^{k-1}\big)-u\| \leq \|x^k-u\| + \alpha_k\frac{\theta_k}{\alpha_k}\|x^k-x^{k-1}\|.$$

Since $\lim\limits_{k\to\infty}\frac{\theta_k}{\alpha_k}\|x^k-x^{k-1}\| = 0$, there exists $M_1 > 0$ such that $\frac{\theta_k}{\alpha_k}\|x^k-x^{k-1}\| \leq M_1$ for all $k \geq 1$. This implies that

$$\|w^k-u\| \leq \|x^k-u\| + \alpha_k M_1. \tag{49}$$

It follows from (47), (48), and (49) that

$$
\begin{aligned}
\|x^{k+1}-u\| &\le \alpha_k\|u\| + (1-\alpha_k)\Big[\|x^k-u\| + \alpha_k M_1\Big] \\
&\le (1-\alpha_k)\|x^k-u\| + \alpha_k\big(\|u\|+M_1\big) \\
&\le \max\{\|x^k-u\|, \|u\|+M_1\} \\
&\dots \\
&\le \max\{\|x^0-u\|, \|u\|+M_1\}.
\end{aligned}
$$

This implies that the sequence $\{x^k\}$ is bounded.

(ii) From the definition of w^k, we have

$$
\begin{aligned}
\|w^k-u\|^2 &\le \|x^k-u\|^2 + \theta_k^2\|x^k-x^{k-1}\|^2 + 2\theta_k\|x^k-u\|\|x^k-x^{k-1}\| \\
&\le \|x^k-u\|^2 + \theta_k\|x^k-x^{k-1}\|M_2,
\end{aligned}
\tag{50}
$$

where $M_2 = \sup_{k\ge1}\big\{2\|x^k-u\| + \alpha_k M_1\big\}$. Since $P_{\mathcal{C}_{k-1}}$ and $P_{\mathcal{C}_k}$ are both nonexpansive, Steps 2, 3 in Algorithm 2, (47), and (50), we have

$$
\begin{aligned}
\|x^{k+1}-u\|^2 &= \|P_{\mathcal{C}_k}P_{\mathcal{C}_{k-1}}z^k - P_{\mathcal{C}_k}P_{\mathcal{C}_{k-1}}u\|^2 \le \|z^k-u\|^2 \\
&= \|(1-\alpha_k)(y^k-u) + \alpha_k(-u)\|^2 \\
&\le \alpha_k^2\|u\|^2 + (1-\alpha_k)^2\|w^k-u\|^2 + 2\alpha_k(1-\alpha_k)\langle w^k-u, -u\rangle \\
&\quad + 2\alpha_k(1-\alpha_k)\gamma_k\Big[\mu\Big\|\sum_{j=1}^N T_{j,k}(w^k)\Big\| + (1-\mu)\Big\|\sum_{j=1}^N T_{j,k-1}(w^k)\Big\|\Big]\|u\| \\
&\le (1-\alpha_k)\|x^k-u\|^2 + \alpha_k\Big[\alpha_k\|u\|^2 + 2(1-\alpha_k)\langle w^k-u, -u\rangle \\
&\quad + (1-\alpha_k)\frac{\theta_k}{\alpha_k}\|x^k-x^{k-1}\|M_2 \\
&\quad + 2(1-\alpha_k)\gamma_k\Big(\mu\Big\|\sum_{j=1}^N T_{j,k}(w^k)\Big\| + (1-\mu)\Big\|\sum_{j=1}^N T_{j,k-1}(w^k)\Big\|\Big)\|u\|\Big] \\
&\le (1-\alpha_k)\|x^k-u\|^2 + \alpha_k\widehat{b}_k.
\end{aligned}
$$

Thus, (45) holds.

Using (44) and (50), and using the same arguments as in the proof of (27), we obtain

$$
\begin{aligned}
\|x^{k+1}-u\|^2 &\le \|x^k-u\|^2 - \varepsilon\frac{\big[\mu\sum_{j=1}^N\|S_{j,k}(w^k)\|^2 + (1-\mu)\sum_{j=1}^N\|S_{j,k-1}(w^k)\|^2\big]^2}{\mu\|\sum_{j=1}^N T_{j,k}(w^k)\|^2 + (1-\mu)\|\sum_{j=1}^N T_{j,k-1}(w^k)\|^2} \\
&\quad - \|(I-P_{\mathcal{C}_{k-1}})z^k\|^2 - \|(I-P_{\mathcal{C}_k})P_{\mathcal{C}_{k-1}}z^k\|^2 \\
&\quad + \alpha_k\|u\|^2 + (1-\alpha_k)\theta_k\|x^k-x^{k-1}\|M_2 \\
&= \|x^k-u\|^2 - \widehat{c}_k + \alpha_k\|u\|^2 + (1-\alpha_k)\theta_k\|x^k-x^{k-1}\|M_2.
\end{aligned}
$$

Thus, (46) holds. □

Theorem 10 *Let $\{x^k\}$ be a sequence generated by Algorithm 2. Then the sequence $\{x^k\}$ converges strongly to an element $u^* \in \Omega$, where $u^* = P_\Omega(0)$.*

Proof It follows from Proposition 9 with u replaced by u^* that

$$\|x^{k+1} - u^*\|^2 \le (1-\alpha_k)\|x^k - u^*\|^2 + \alpha_k \widehat{b}_k^*,$$
$$\|x^{k+1} - u^*\|^2 \le \|x^k - u^*\|^2 - \widehat{c}_k^* + \alpha_k\|u^*\|^2 + (1-\alpha_k)\theta_k\|x^k - x^{k-1}\|M_2^*,$$

where

$$\widehat{b}_k^* = \alpha_k\|u^*\|^2 + 2(1-\alpha_k)\langle w^k - u^*, -u^*\rangle + (1-\alpha_k)\theta_k\|x^k - x^{k-1}\|M_2^*$$
$$+ 2(1-\alpha_k)\gamma_k\Big[\mu\Big\|\sum_{j=1}^N T_{j,k}(w^k)\Big\| + (1-\mu)\Big\|\sum_{j=1}^N T_{j,k-1}(w^k)\Big\|\Big]\|u^*\|,$$
$$\widehat{c}_k^* = \varepsilon\frac{\big[\mu\sum_{j=1}^N\|S_{j,k}(w^k)\|^2 + (1-\mu)\sum_{j=1}^N\|S_{j,k-1}(w^k)\|^2\big]^2}{\mu\|\sum_{j=1}^N T_{j,k}(w^k)\|^2 + (1-\mu)\|\sum_{j=1}^N T_{j,k-1}(w^k)\|^2}$$
$$+ \|(I - P_{\mathcal{C}_{k-1}})z^k\|^2 + \|(I - P_{\mathcal{C}_k})P_{\mathcal{C}_{k-1}}z^k\|^2,$$
$$M_2^* = \sup_{k\ge1}\{2\|x^k - u^*\| + \alpha_k M_1\}.$$

From the definition of w^k and Remark 3, we get

$$\|w^k - x^k\| = \Big\|\alpha_k\frac{\theta_k}{\alpha_k}(x^k - x^{k-1})\Big\| \to 0 \text{ as } k\to\infty. \tag{51}$$

It follows from (4) and Remark 3 that $\alpha_k \in (0,1)$, $\sum_{k=1}^\infty \alpha_k = \infty$, and $\alpha_k\|u^*\|^2 + (1-\alpha_k)\alpha_k\frac{\theta_k}{\alpha_k}\|x^k - x^{k-1}\|M_2^* \to 0$ as $k\to\infty$.

Using (51) and using the same arguments as in the proof of Theorem 6, we obtain, for every subsequence $\{k_l\}$ of $\{k\}$,

$$\lim_{l\to\infty}\widehat{c}_{k_l}^* = 0 \text{ implies } \limsup_{l\to\infty}\widehat{b}_{k_l}^* \le 0.$$

Let $s_k = \|x^k - u^*\|^2$, from Lemma 2, we conclude that $\|x^k - u^*\| \to 0$ as $k\to\infty$, that is, $x^k \to u^* = P_\Omega(0)$ as $k\to\infty$. □

Remark 4 We present the advantages of Algorithms 1, 2 and Theorems 6, 10 compared with the recent results.

1. Compared with Theorem 3.4 in [14] when $u = 0$, Algorithm (4.10) in [16] when $f = 0$, and Algorithm 1 and Theorem 3.1 in [9] when $F = I$, our iterative schemes in Algorithms 1 and 2 are based on replacing the projections $P_{\mathcal{C}}$ and $P_{\mathcal{Q}_j}$ with that

to the half-spaces. So, they are easily calculated. Furthermore, our iterative scheme in Algorithm 2 is embedded with an inertial effect, which allows it to accelerate the convergence speed of the algorithm.

2. In our Algorithms 1, 2, unlike Theorem 3.4 in [14], the step size is selected in such a way that its implementation does not require any prior information of the norm of the operator $\mathcal{F}_j$. Furthermore, compared with Algorithm (4.10) in [16] and Algorithm 1 in [9], the step size we constructed is larger, while a larger step size often means faster convergence speed.

4 Some Corollaries

If $N = 1$, then the SFPMOS reduces to the SFP. In what follows, from Algorithms 1, 2, and Theorems 6, 10, we get some corollaries to SFP. The set $\mathcal{Q}$ is given by

$$\mathcal{Q} = \{y \in \mathcal{H}_1 \mid h(y) \leq 0\},$$

where $h : \mathcal{H}_1 \to \mathbb{R}$ is convex function on $\mathcal{H}_1$. Define a set at point x^k by

$$\mathcal{Q}_k = \{y \in \mathcal{H}_1 \mid h(\mathcal{F}x^k) + \langle \eta_k, y - \mathcal{F}x^k \rangle \leq 0\},$$

where $\eta_k \in \partial h(\mathcal{F}x^k)$. For each $x \in \mathcal{H}$ and for all $k \geq 1$, define

$$T_k(x) = \mathcal{F}^*(I - P_{\mathcal{Q}_k})\mathcal{F}(x) \text{ and } S_k(x) = (I - P_{\mathcal{Q}_k})\mathcal{F}(x).$$

Corollary 11 *Suppose that $g : \mathcal{H} \to \mathbb{R}$ and $h : \mathcal{H}_1 \to \mathbb{R}$ are convex, subdifferentiable and w-lsc functions on $\mathcal{H}$ and $\mathcal{H}_1$, respectively; for any $x \in \mathcal{H}$ and $y \in \mathcal{H}_1$, at least one subgradient $\xi \in \partial g(x)$ and $\eta \in \partial h(y)$ can be calculated; the subdifferential operators ∂g and ∂h are bounded on bounded sets. Assume that the sequences $\{\alpha_k\}$ and $\{\rho_k\}$ satisfy the conditions (4) and (9). For given $x^0, x^1 \in \mathcal{H}$, let $\{x^k\}$ be the sequence generated by*

$$\begin{cases} y^k = x^k - \gamma_k\big[\mu T_k(x^k) + (1-\mu)T_{k-1}(x^k)\big], \\ z^k = (1-\alpha_k)y^k, \\ x^{k+1} = P_{\mathcal{C}_k} P_{\mathcal{C}_{k-1}} z^k, \end{cases} \tag{52}$$

where $\mu \in (0, 1)$, the step size γ_k is defined by (11).

Then the sequence $\{x^k\}$ converges strongly to $u^* \in \Omega_{\text{SFP}}$, where $u^* = P_{\Omega_{\text{SFP}}}(0)$, provided that the solution set $\Omega_{\text{SFP}} = \{u^* \in \mathcal{C} \mid \mathcal{F}u^* \in \mathcal{Q}\}$ of the SFP *is nonempty.*

Corollary 12 *Suppose that all the conditions in Corollary 11 are satisfied. Assume that the sequence $\{\eta_k\}$ satisfies $\lim_{k\to\infty} \frac{\eta_k}{\alpha_k} = 0$. For given $x^0, x^1 \in \mathcal{H}$, let $\{x^k\}$ be the* *sequence generated by*

$$\begin{cases} w^k = x^k + \theta_k \left(x^k - x^{k-1}\right), \\ y^k = w^k - \gamma_k\left[\mu T_k(w^k) + (1-\mu)T_{k-1}(w^k)\right], \\ z^k = (1-\alpha_k)y^k, \\ x^{k+1} = P_{\mathcal{C}_k} P_{\mathcal{C}_{k-1}} z^k, \end{cases} \tag{53}$$

where $\mu \in (0,1)$, θ_k *is defined by (42) and* γ_k *is defined by (11) with* x^k *replaced by* w^k.

Then the sequence $\{x^k\}$ *converges strongly to* $u^* \in \Omega_{\text{SFP}}$, *where* $u^* = P_{\Omega_{\text{SFP}}}(0)$, *provided that the solution set* Ω_{SFP} *of the* SFP *is nonempty.*

Remark 5 We present the advantages of Corollaries 11, 12 compared with the recent results.

1. Compared with Algorithm 4.1 and Theorem 4.3 of Yu & Wang [19], Algorithm 3.1 and Theorem 3.5 of López et al. [13], and Algorithm (2.3) and Theorem 1 of Yang [18], our iterative scheme in Corollary 12 is embedded with an inertial effect, which allows it to accelerate the convergence speed of the algorithm. Furthermore, our algorithms in Corollaries 11 and 12 give strong convergence, while the algorithms of Yu & Wang [19], López et al. [13] give only weak convergence.
2. In Corollaries 11 and 12, unlike Algorithm (2.3) of Yang in [18], the step size is selected in such a way that its implementation does not require any prior information of the norm of the operator $\mathcal{F}$.

5 Numerical Experiments

In this section, we perform some computational tests that occur in finite-dimensional spaces to verify the proposed algorithm. Also, we considered an application in compressed sensing and provided some comparisons to the algorithm (6) of Yang [18], the algorithm (7) of López et al. [13], and the algorithm (10) of Yu & Wang [19]. The code is implemented in MATLAB R2019a on a Intel(R) Core(TM) i5-9300 H CPU @2.40GHz computer with RAM 16GB.

Example 13 (Numerical example to (12)) We consider the SFPMOS (12), where the closed convex sets $\mathcal{C}$ and $\mathcal{Q}_j$ are given by (A2) with the functions $g : \mathbb{R}^3 \to \mathbb{R}$, $h_j : \mathbb{R}^{3+j} \to \mathbb{R}$, $j = 1, \dots, N$, are defined by

$$g(x) = x_1^2 + x_2 + 2x_3^2 - 1,\ x \in \mathbb{R}^3 \text{ and } h_j(y) = \|y\|_1 - j,\ y \in \mathbb{R}^{3+j},$$

the operator $\mathcal{F}_j : \mathbb{R}^3 \to \mathbb{R}^{3+j}$, $j = 1, \dots, N$, is defined by

$$\mathcal{F}_j(x) = (x_1 - 3x_3, -3x_2 + 4x_3, 3x_1 - 4x_2, (3+1)x_1, \dots, (3+j)x_1),\ x \in \mathbb{R}^3.$$

It is easy to see that g and h_j are convex functions, and $\mathcal{F}_j$ is a bounded linear operator. Since $0 \in \Omega$, $\Omega \neq \emptyset$ and $P_\Omega(0) = 0$. So such an SFPMOS can be solved

Table 1 Numerical results for Algorithms 1 and 2 with different initial points

Initial points	Algorithm 1		Algorithm 2	
	Iter	CPU (s)	Iter	CPU (s)
$x^0 = (1, 2, 3), x^1 = (-3, -4, -5)$	2342	1.3147	74	0.0423
$x^0 = (100, 150, 200), x^1 = (-50, -40, 100)$	6574	3.7729	281	0.2158

Table 2 Numerical results for Algorithms 1 and 2 with different α_k

	Algorithm 1		Algorithm 2	
α_k	Iter	CPU (s)	Iter	CPU (s)
$\frac{1}{(k+1)^{0.4}}$	13	0.0294	12	0.0279
$\frac{1}{(k+1)^{0.6}}$	30	0.0407	17	0.0317
$\frac{1}{(k+1)^{0.8}}$	105	0.0926	24	0.0161

by the proposed algorithms. Throughout the experiment, the parameters used in these algorithms are set with $N = 30, \theta = \mu = 0.5, \alpha_k = \frac{1}{k+1}, \eta_k = \frac{1}{(k+1)^{1.01}}, \rho_k = 1.5$ for all $k \geq 1$, err $= 10^{-3}$. The stopping criteria is that $\|x^k - P_\Omega(0)\| <$ err. The results are listed in Table 1 using different initial points. In Table 1, Iter. denotes the number of iterations, and CPU denotes the computing time.

In Table 2 we present results of Algorithms 1 and 2 with the initial point $x^0 = (1, 2, 3)$, $x^1 = (-3, -4, -5)$ and different choices of α_k. The stopping condition is set as $\|x^k - P_\Omega(0)\| < 10^{-3}$.

Example 14 (Application in signal processing) We aim to apply our main result to signal recovery in compressed sensing. It is modeled as the following equation:

$$y = \mathcal{F}x + \epsilon, \tag{54}$$

where $x \in \mathbb{R}^L$ is a recovered vector containing m components which are nonzero, $y \in \mathbb{R}^K$ is an observed data, ϵ is the noise and $\mathcal{F}$ is a bounded and linear mapping from $\mathbb{R}^L$ to $\mathbb{R}^K$. It is known that (54) is equivalent to the following constrained minimization problem (see [10])

$$\min_{x \in \mathbb{R}^L} \frac{1}{2}\|y - \mathcal{F}x\|_2^2 \text{ subject to } \|x\|_1 \leq r, \tag{55}$$

where r is a positive parameter, $\|.\|_2$ is the Euclidean norm, and $\|.\|_1$ is the ℓ_1-norm defined by $\|x\|_1 = \sum_{k=1}^{L} |x_k|$. The CQ algorithm can be applied to approximate solutions to (55) because it is a particular case of the SFP where $\mathcal{C} = \{x \in \mathbb{R}^L \mid \|x\|_1 \leq r\}$ and $\mathcal{Q} = \{y\}$. By using the relaxed CQ method, we define the convex function $g(x) = \|x\|_1 - r$.

Table 3 Performance of the four algorithms with $L = 1024$ and $K = 512$

m-sparse	Algorithm	MSE $< 10^{-4}$		MSE $< 10^{-5}$	
		Iter	CPU (s)	Iter	CPU (s)
$m = 10$	Alg. 4	21	0.0214	30	0.0271
	Yang's Alg	50	0.2040	74	0.2813
	López's Alg	37	0.1440	56	0.2268
	Yu & Wang's Alg	25	0.0258	39	0.0467
$m = 20$	Alg. 4	29	0.0251	43	0.0439
	Yang's Alg	73	0.2938	115	0.4879
	López's Alg	52	0.1818	76	0.3153
	Yu & Wang's Alg	33	0.0382	54	0.0658
$m = 30$	Alg. 4	33	0.0239	44	0.0502
	Yang's Alg	98	0.3813	164	0.6325
	López's Alg	66	0.2546	107	0.4437
	Yu & Wang's Alg	39	0.0379	60	0.0748

In our numerical experiment, we generate the sparse vector x^* by the uniform distribution in $[-2, 2]$ with m nonzero elements. We generate $\mathcal{F}$ by the normal distribution with mean zero and one variance. The observed signal y is generated by white Gaussian noise with signal-to-noise ratio SNR $= 40$.

We aim to compare our algorithm (53) to the algorithm (6) of Yang [18], the algorithm (7) of López et al. [13], and the algorithm (10) of Yu & Wang [19]. For convenience, we denote algorithm (53), the algorithm of Yang [18], the algorithm of López et al. [13], and the algorithm of Yu & Wang [19] by Alg. 4, Yang's Alg., López's Alg., and Yu & Wang's Alg., respectively. Set $r = m$, $x^0 = (0, 0, \dots, 0)^\top$, $x^1 = (1, 1, \dots, 1)^\top \in \mathbb{R}^L$. The parameters of these methods are chosen as follows:

- In Alg. 4 (53), we choose $\theta = 0.5$, $\mu = 0.9$, $\alpha_k = \frac{1}{100k+1}$, $\eta_k = \frac{1}{(k+1)^{1.01}}$ and $\rho_k = 1.5$;
- In Yang's Alg., we choose $\gamma_k = \frac{1.5}{\|\mathcal{F}\|^2}$;
- In López's Alg., we choose $\rho_k = 1.5$;
- In Yu & Wang's Alg., we choose $\mu = 0.9$ and $\rho_k = 1.5$.

To measure the recovery accuracy, we use the following mean square error method:

$$\text{MSE} = \frac{1}{L}\|x^k - x^*\|^2,$$

where x^k is the recovered signal at k-th iteration.

In Table 3, we observe that the performance of Alg. 4 (53) is better than other algorithms in terms of CPU time and the number of iterations as the spikes of sparse vectors are varied from 10 to 30.

We next provide illustrations of the original signal and the recovered signal in case $L = 1024$, $K = 512$, $m = 30$, and MSE $= 1e$-5 (Fig. 1).

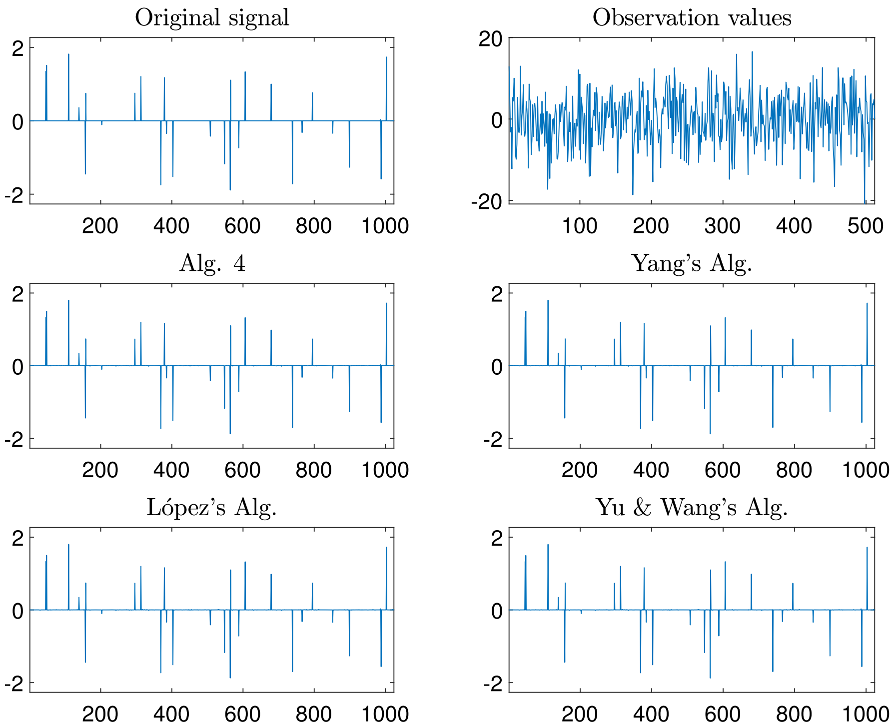

Fig. 1 The original signal, observed signal, restored signal by Alg. 4 (53) and the algorithms in [13, 18, 19] with $L = 1024$, $K = 512$ and $m = 30$

Finally, we give some graphs of MSE plotted with iterations and time. As we can see from Fig. 2, our relaxed CQ algorithm Alg. 4 (53) demonstrates better performance compared with Yang's algorithm, López's algorithm, and Yu & Wang's algorithm, in terms of CPU time and the number of iterations.

6 Conclusion

The paper has proposed two relaxed methods (Algorithm 1 and Algorithm 2) for solving the split feasibility problem with multiple output sets in real Hilbert spaces. Our algorithms have two advantages, one is that the methods are based on replacing the projection onto the nonempty closed convex subsets with that to the half-spaces and the other is that the step size does not depend on the operator norm. Furthermore, in Algorithm 2, we use an inertial step, which increases the convergence rate. We gave two strongly convergence theorems under suitable conditions. As a special case, we have also provided two strong convergent algorithms to approximate the solution to the split feasibility problem. We applied the result to signal recovery and provided a comparison with other methods. Results showed that our method is more efficient than other methods in terms of iteration and CPU time.

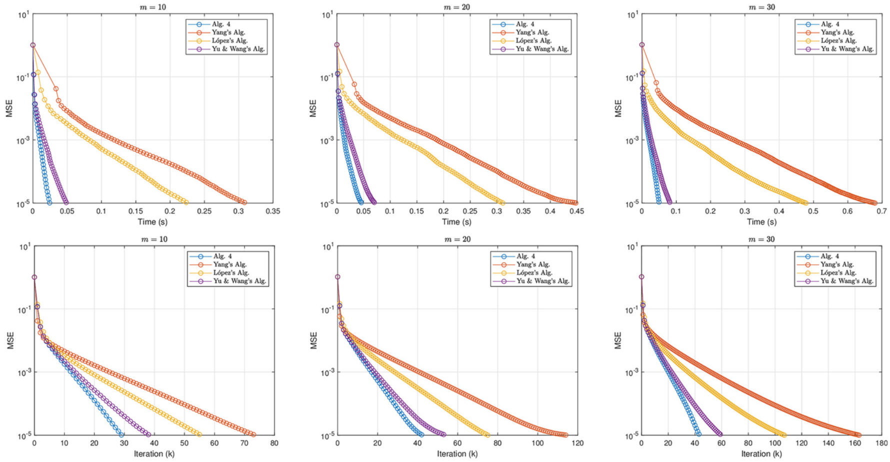

Fig. 2 MSE with number of iterations and time when $L = 1024$, $K = 512$ with different m

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Declarations

Conflict of interest The authors declare that there is no conflict of interest regarding the publication of this manuscript.

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